

IBM i 5250 Terminal Emulator for
Development Support
(with MCP Server support)

IBM 5250 MCP

User Guide

Important Safety Information

Before using this product, be sure to read this "Important Safety Information" carefully and use the product correctly.

Caution	Incorrect handling may affect business data or resources on the IBM i.
Prohibited	Operations that must never be performed.
Note	Useful tips and reference information.



Confirm Your Operations

This software communicates with IBM i (formerly AS/400, iSeries). Before running any operation that may unintentionally change data on the server (for example, delete commands or system value updates), always confirm its content.



About Passwords

Do not leave the device or share the screen while a password is entered on screen. Handle passwords in accordance with your organization's operational rules.



About the Screen Examples in This Guide

The screens shown in this guide were captured in a test environment. The display may differ from your own environment.

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Chapter 1

Introduction

Introduces the product overview and how to read this guide.

1.1 Product Overview

IBM 5250 MCP is development-support software for operating, in a web browser, 5250 terminal screens that run on IBM i (formerly AS/400, iSeries). It also works as a Model Context Protocol (MCP) server, supporting automated operation by generative AI agents (such as Claude).

1. **Operate in a web browser** No dedicated client needs to be installed. You can operate 5250 screens from your own web browser.
2. **Japanese display support** Correctly displays screens that mix DBCS (double-byte characters) and SBCS (single-byte characters).
3. **Automatic input feature** You can save a frequently performed sequence of operations as an "AUTOTYP script" and reproduce it with a single click.
4. **MCP server support** Generative AI agents can invoke and automatically operate 5250 screens. This is useful for test automation and for embedding into operational batch processing.

Supported IBM i Versions

This product implements the TN5250E protocol (compliant with RFC 4777 / RFC 1205). You can use it without any special configuration on a typical IBM i 7.x system.

1.2 Structure of This Guide

Chapter	Main Content
Chapter 1	Introduces the product overview and the structure of this guide.
Chapter 2	Explains the names and roles of the parts of the screen.
Chapter 3	Explains the procedure for starting a connection and signing on.
Chapter 4	Explains basic operations such as entering text, function keys, and copy and paste.
Chapter 5	Explains useful features such as AUTOTYP, operation recording, and display switching.
Chapter 6	Guides you through what to check when something is not working correctly.
Chapter 7	Explains the technical terms used in this guide.
Chapter 8	A list of keyboard shortcuts.

For Administrators

For administrator-oriented content such as creating connection configurations, developing AUTOTYP scripts, and handling print spools, please refer to the separate "IBM 5250 MCP Administrator Guide".

Chapter 2

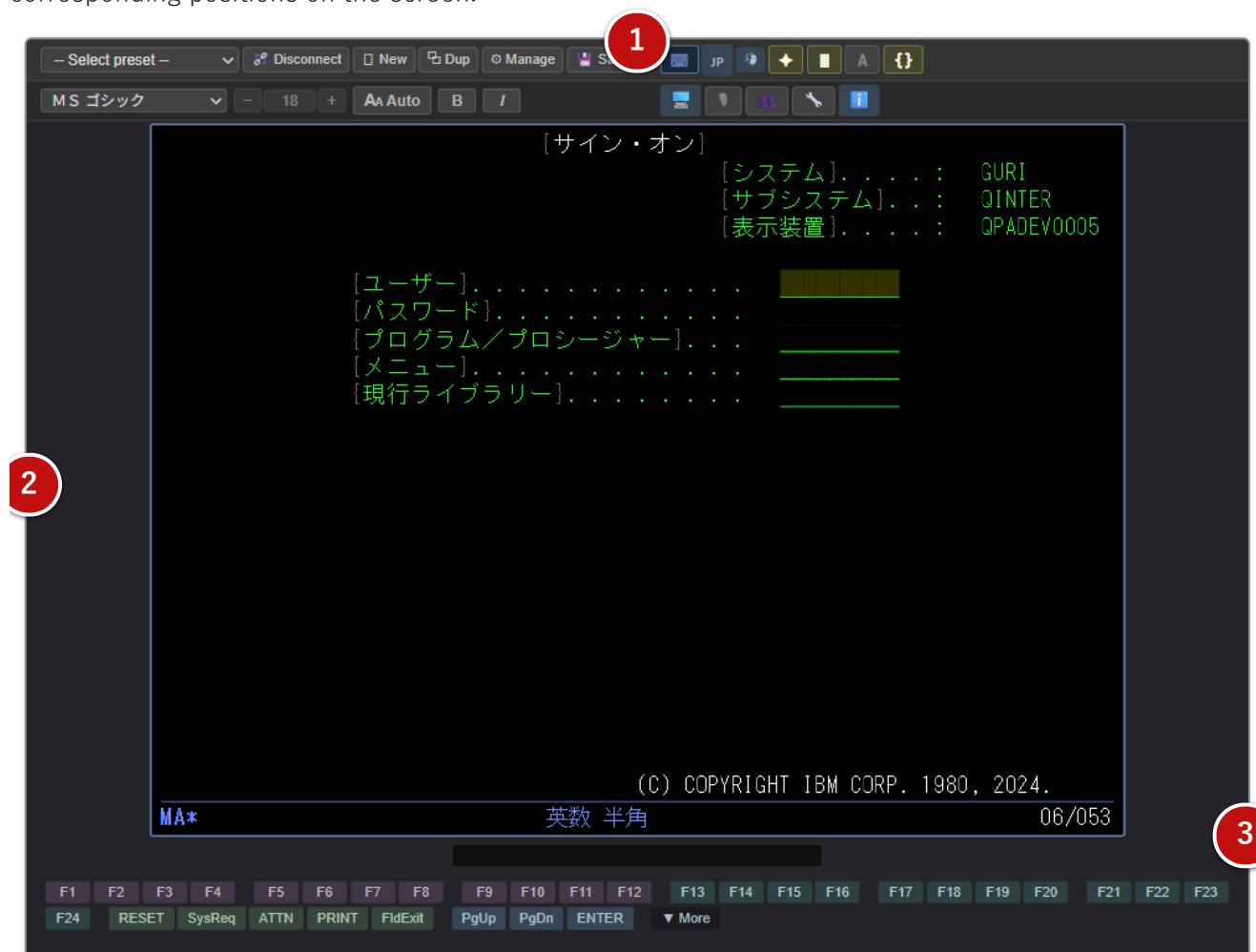
Names of Parts

Explains the names and roles of the parts of the screen.

Names of Parts

2.1 The Whole Screen

The screen of this product consists of the following three areas. Check the numbers ①–③ and their corresponding positions on the screen.



- 1 Toolbar** There are two toolbar rows, upper and lower. The first row contains connection management, configuration editing, and saving settings; the second row contains the display font and size settings. Buttons for the input target (KBD ON/OFF) and display switching are also included in this area. Function keys and AUTOTYP operation buttons are also placed around the screen.
- 2 5250 Screen** Displays the screen sent from IBM i. The "input fields" for entering text are also shown here.
- 3 Operator Information Area (OIA)** Displays the keyboard input state, cursor position, communication state, and so on. For details, see section 2.3.

2.2 Toolbar Description

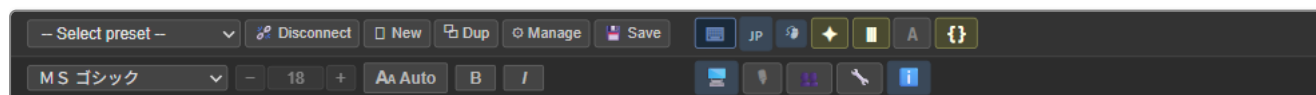
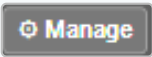
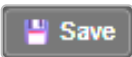


Figure 2.2-1 Upper Toolbar

Each button is described below (the left column shows the actual button appearance).

Button	Description
First Row — Connection / Configuration	
 Connect	Disconnect / Connect. Disconnects the current session. While disconnected, the button changes to Connect. When connecting, the connection begins in the current window; no new window is opened.
 New	New. Opens a new, unconnected window. Use this when you want to use multiple 5250 sessions at the same time.
 Dup	Duplicate. Opens a new window using the same connection configuration as the current session (the connection is also made immediately). When not connected, this behaves the same as "New" and opens an unconnected window.
 Manage	Manage. Displays a dialog for adding, editing, and deleting connection destination configurations. The default display font and size can also be specified for each configuration.
 Save	Save. Saves the current display font, size, bold, italic, and other settings to the connected configuration. The next time you connect with the same configuration, those settings are restored.
	Keyboard input target. Switches the keyboard input target between the 5250 screen (on) and toolbar operations (off). You can also switch with  +  .
	Language. Switches the display language of the viewer UI (Japanese / English). Switching it also changes the default font and default host code page used when creating a new connection configuration (Japanese: MS Gothic / cp1399, English: Consolas / cp037). Already-saved configurations are not affected.
	Wheel settings. Opens a settings dialog for screen scrolling with the mouse wheel.
	Input highlight. Toggles the yellow highlighting of input fields (on / off).
	Column separator. Switches the column separator display between "lines" and "dots".
	A / ｱ. Switches the SBCS code page (DBCS sessions only). A = lowercase Latin (cp1027), ｱ = Katakana (cp290)
	{ } / ｼ. Shows or hides shift characters (SO/SI) (DBCS sessions only).
Second Row — Display Font	
Font selection	Selects the display font for the 5250 screen. At the top of the list are monospace fonts (recommended), and below them all the fonts installed on the OS are listed. In Japanese (DBCS) sessions, a Japanese font that can correctly display kanji is selected by default.
– / Size / +	Changes the font size. In automatic mode, the size currently in use is shown (cannot be changed); in fixed mode, you can enter a value directly. – / + increase or decrease by 1 px at a time.
 Aa Auto	Auto / Fixed. Switches the size adjustment method. Auto automatically adjusts the size to fit the 5250 screen within the window. Fixed locks the size to the specified value and adjusts the window size so that the 5250 screen fits exactly.
	Bold (B). Toggles bold display on / off.
	Italic (I). Toggles italic display on / off.
	Window mode toggle. Switches the viewer window between an "app window" (a dedicated window without an address bar) and a normal browser window. The button shows the icon of the mode it will switch to (🌐 = to a normal window / 🖥️ = to an app)

	window). Clicking it closes the current window, reopens it in the other mode, and also changes the default for the next startup.
	Microphone. Click to start voice input; it stops automatically when you finish speaking (click again while recording to stop midway). For details, see section 5.5.
	Exclusive / Shared. Switches the input mode. Exclusive (the default at startup) means that even if the same connection is opened in another browser or PC, only the one person holding exclusive control can enter input. With Shared , everyone can enter input, and multiple people can operate at the same time. While someone holds exclusive control, other browsers cannot switch to exclusive (the button appears disabled). For details, see section 2.4.
	Hide all toolbars. Hides all toolbars (the two upper toolbar rows, the notification area, and the function key toolbar) and displays only the 5250 screen (the OIA remains). This is a hide-only button. To show them again, click the gray area outside the 5250 screen (because the button itself also disappears while hidden). When hidden, the keyboard input target switches to the 5250 screen (Keyboard on), and the 5250 screen font size is readjusted to fit the freed-up area. You can specify the initial state per connection configuration ("Toolbar initially hidden" in the Manage dialog), and it is applied when connecting.
	Help (i). Opens the help in a normal browser window. It shows the product version and build date, the End-User License Agreement (EULA), third-party notices, this User Guide, and the Admin Guide. For details, see section 5.6.

About the Default Values of the Display Switching Buttons

For the four display switches "Input highlight", "Column separator", "A / ｱ", and "{ } / . .", the default values are saved as a single set for the entire viewer (= not per connection configuration). Once set, they are retained even after reloading the browser.

2.3 Operator Information Area (OIA)

The status line at the bottom edge of the 5250 screen (the Operator Information Area, abbreviated as **OIA**) displays the current keyboard state and communication state.



Figure 2.3-1 Close-up of the Operator Information Area (OIA)

Display	Meaning
MA* (column 1)	Indicates that you are connected to IBM i.
X II (column 9)	The keyboard is temporarily locked. This product is processing the previous operation (such as printing). You can release it with the RESET key.
X SYSTEM (column 9)	Waiting for a response from the server. Please wait a moment.
COMM659 (column 19)	Communication has been disconnected. Please reconnect. It disappears automatically when communication is restored.
Alphanumeric / Kana (column 26)	Indicates the current input mode (DBCS sessions only).
A (column 43)	Indicates that Caps Lock is on.
^ (column 52)	Indicates insert mode. It appears whenever insert mode is on.
18/007 (column 75)	Indicates the current cursor position (row 18 / column 7). It is shown as a combination of a 2-digit row and a 3-digit column.

Cursor Display in Insert Mode

In insert mode, the cursor changes from a "tall block (overwrite mode)" to a "lower-half block (insert mode)". Pressing the **Insert** key toggles it (on / off), and the **RESET** key also returns it to overwrite mode (this is not a toggle; it only releases insert mode).

In insert mode, "^" also appears at OIA column 52.

2.4 Exclusive and Shared Input

You can open the same connection in multiple browsers or PCs at the same time. In this case, the "input mode" controls who can enter input. Switch with the  /  button on the toolbar (in either mode, everyone who has it open can view the screen).

Mode	Description
 Exclusive (default at startup)	Only the one person holding exclusive control can enter input. Even if the same connection is opened in another browser or PC, others cannot enter text, send keys, or move the cursor.
 Shared	Everyone who has the connection open can enter input. Use this when you want multiple people to operate at the same time. Input is processed in the order it arrives.

 While someone holds exclusive control, the  button in other browsers appears disabled and cannot be switched to exclusive. When the person who held exclusive control switches to  Shared or closes the connection, another browser becomes able to take exclusive control.

Chapter 3

Before You Begin

Explains the procedure for starting a connection and signing on.

3.1 Starting a Connection

The procedure for starting a connection is as follows.

1

Open the product screen in your browser

Open the address provided in advance by your administrator (for example,

`http://localhost:5250/`) in a web browser. If you started the product directly, an unconnected screen opens automatically.

i When using an AI agent (MCP)

If you asked an AI agent such as Claude to connect, this step is not needed. When the connection completes, a window showing that session opens automatically (one connection = one window).

2

Select a connection configuration

From the "-- Select configuration --" drop-down at the top of the toolbar, select the connection destination you want to use. Selecting it starts the connection immediately.

i If the Connection Configuration Is Not in the List

If the connection destination you want is not in the list, contact your administrator.

Alternatively, you can add a new one from the "⚙ Manage configurations" button (for details, see the "Administrator Guide").

3

The sign-on screen appears

When the connection succeeds, the IBM i sign-on screen appears.

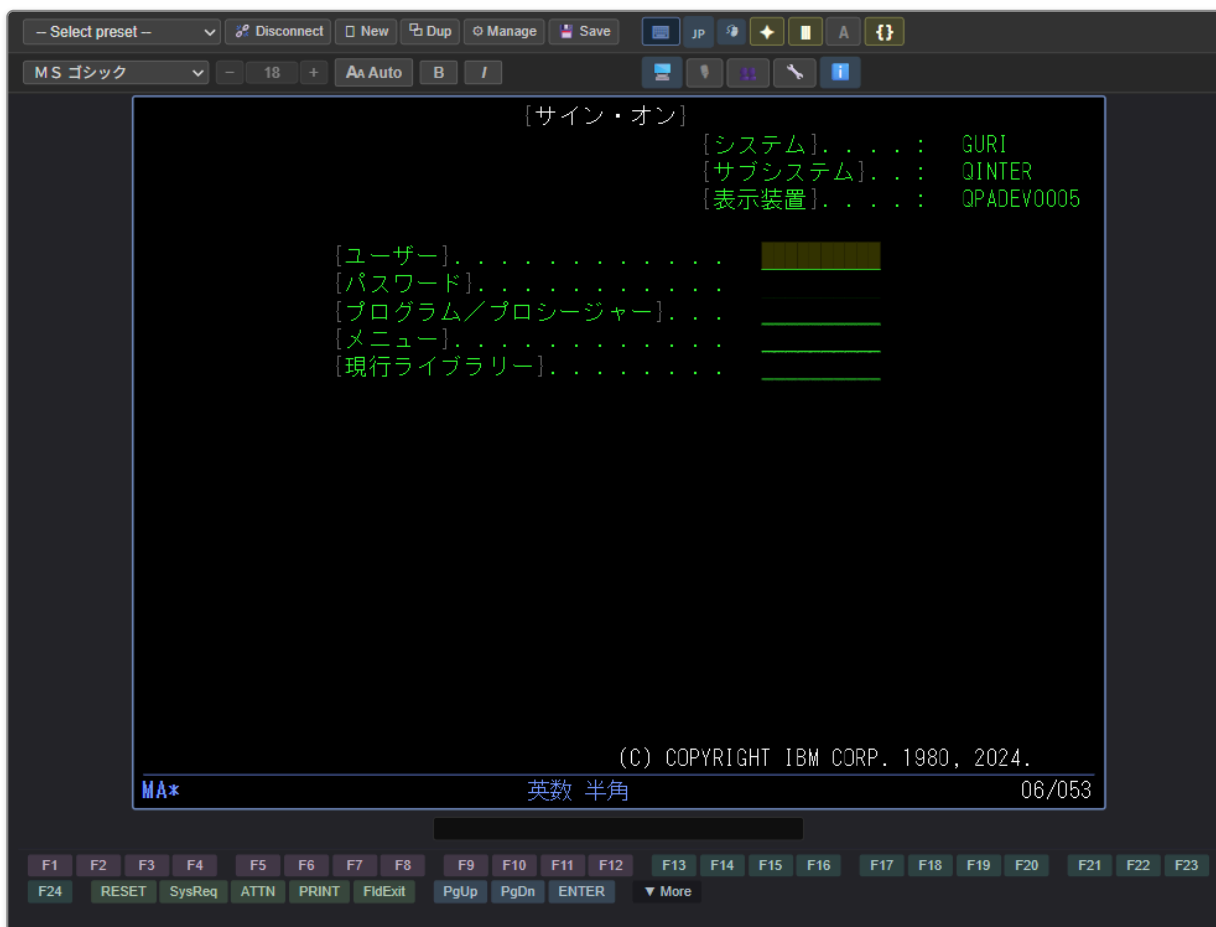


Figure 3.1-1 Sign-on Screen

3.2 Signing On

1

Enter your user name

Enter your user name (the value provided by your administrator) in the "User" field.

2

Move to the password field

Press the key to move the cursor to the "Password" field.

3

Enter your password and press the Enter key

After entering your password, press the **Enter** key. When sign-on succeeds, the IBM i menu screen (or command entry screen) appears.

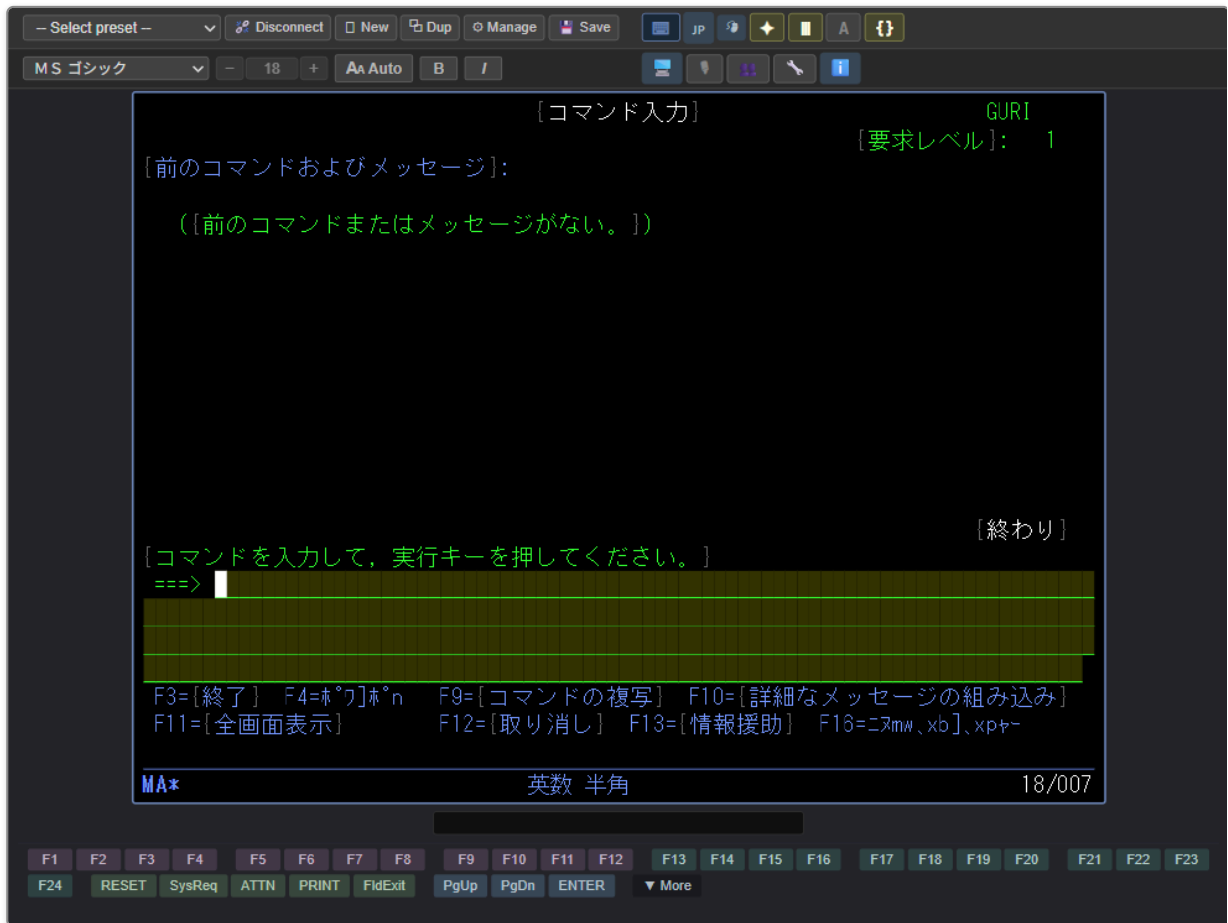


Figure 3.2-1 Command Entry Screen After Sign-on



When Sign-on Fails

Confirm that the user name and password are correct. The account may also be disabled on the IBM i side. If you still cannot sign on after several attempts, contact your administrator.

Chapter 4

Basic Operations

Explains entering text, function keys, copy and paste, and more.

4.1 Entering Text

You can enter text from the keyboard at the position where the cursor is blinking (= an input field).

1 Select an input field

Click the position where you want to enter text with the mouse, or use the **Tab** key to move to the next field. To move in the reverse direction, press **Shift** + **Tab**.

2 Enter text

Enter text from the keyboard. Japanese input (IME) can also be used.

3 Press a confirm key

Pressing the **Enter** key, or the appropriate **F1**–**F24** key, sends the entered content to IBM i.

i Keys Useful for Editing

- **BackSpace**: Moves the cursor one character to the left (the character is not deleted).
- **Delete**: Deletes the character at the cursor position and shifts the text to the right leftward.
- **Insert**: Switches between insert mode (= inserting characters in between) and overwrite mode. In insert mode, "^" appears at OIA column 52.
- **End**: Erases from the cursor position to the end of the line (Erase EOF).

i About Japanese (DBCS) Input Fields

In a field for entering double-byte characters (kanji and kana), the mode automatically switches to "Kana" when the cursor enters (shown in the OIA). The characters that can be entered are determined by the type of field, and if you enter a character that does not match the type, a message appears at the bottom of the screen and the input is rejected.

- **Double-byte-only field**: If you enter alphanumeric characters, "DBCS is required as input." appears.
- **Either-or field**: If the first character entered is alphanumeric, that field becomes alphanumeric-only; if it is a double-byte character, it becomes double-byte-only (mixing is not allowed).

4.2 Function Keys (F1–F24)

In 5250 applications, the F1–F24 function keys are used frequently. The common assignments are as follows.

Key	Function (common example)
F1	Displays help.
F3	Ends the current process.
F4	Displays a list of choices (prompt).
F5	Refreshes the screen to the latest state.
F12	Cancels the process and returns to the previous screen.

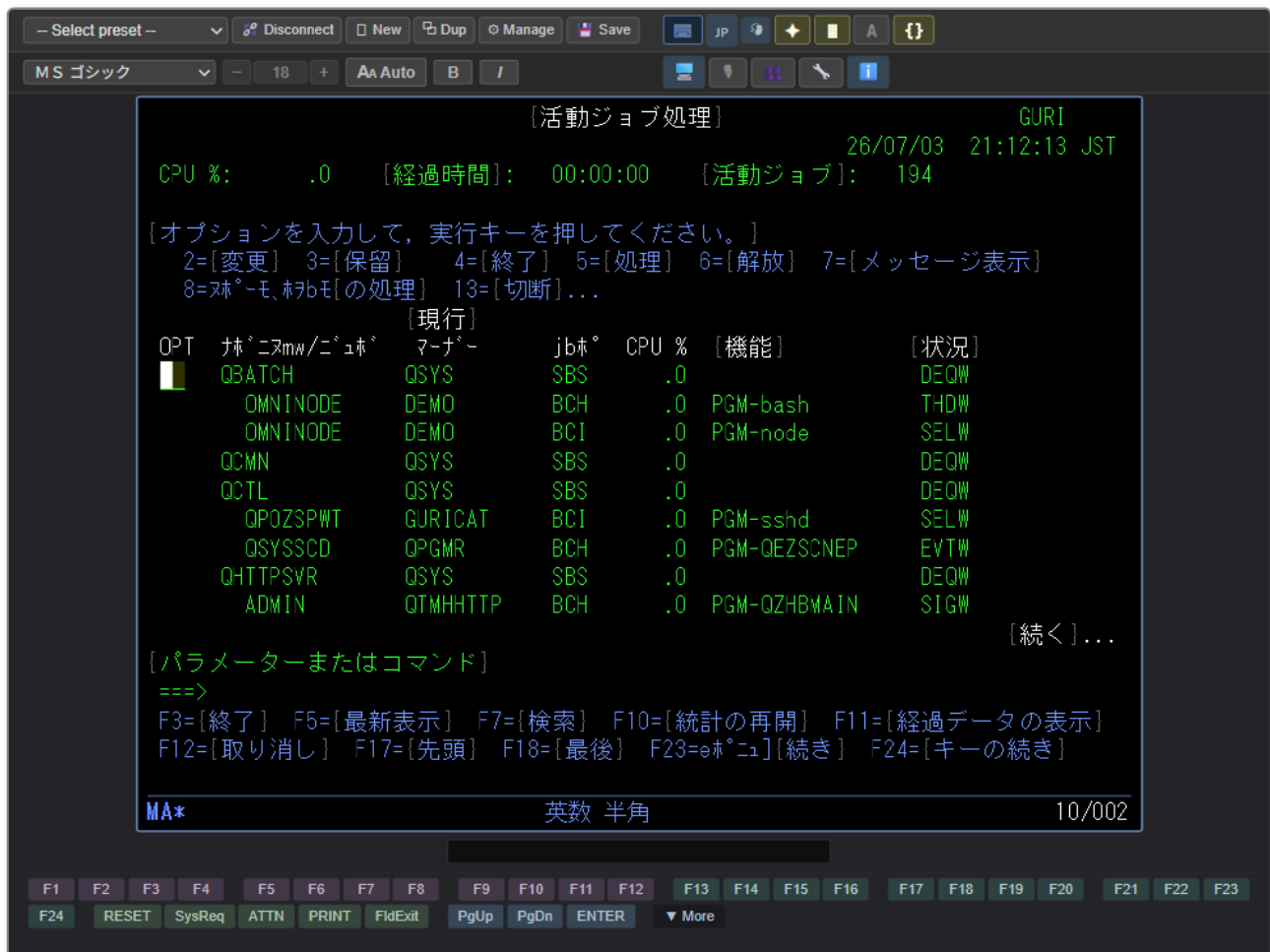


Figure 4.2-1 Example of function key descriptions shown at the bottom of the screen

i How to Enter F13–F24

F13–F24 can also be entered with **Shift** + F1–F12. They can also be operated as on-screen buttons from the **▼ More** button at the bottom.

4.3 Copy and Paste

1 Select a range

Dragging the mouse highlights the range you want to copy in yellow (just drag—you do not need to hold **Ctrl** or **Shift**).

You can also select with the keyboard. Hold **Shift** and move the cursor with the arrow keys (**↑** **↓** **←** **→**) to highlight the range you pass over.

2 Copy

Pressing **Ctrl** + **C** copies the selected range to the clipboard.

3 Paste

Place the cursor in an input field and press **Ctrl** + **V**. Multi-line text with line breaks is also pasted automatically according to the arrangement of the fields.

4 Undo the paste (optional)

Pressing **Ctrl** + **Z** immediately after pasting restores the state before the paste. You can undo only immediately after pasting; you cannot undo after pressing another key.

When the Paste Does Not Fit in the Field

If the text being pasted exceeds the number of columns in the field, it is flowed into the input lines below, aligned to the column at the cursor position (block paste).

Paste Caution

When pasting a password from the clipboard into a password field, always clear the clipboard after pasting.

4.4 Mouse Operations

When you click on the 5250 screen with the mouse, the cursor moves to that position. The following operations are also supported.

Operation	Behavior
Click a normal position	The cursor moves to that position (nothing is sent to the server).
Click a choice or push button	You can directly click on-screen choice buttons or push buttons (supporting the DDS MOUBTN keyword) to select or execute them.
Drag	Selects a range (for copying; see section 4.3).
Click the scroll bar arrows (▲ / ▼)	Scrolls one line at a time.
Click the scroll bar track (other than the arrows)	Scrolls one page at a time.

Chapter 5

Useful Features

Introduces various features that streamline your daily work.

5.1 AUTOTYP (Automatic Input)

AUTOTYP is a mechanism that automatically runs a frequently performed sequence of inputs all at once. For example, you can reproduce everything from "signing on to opening a menu" with a single click.

1 Select a script

From the "-- autotyp --" drop-down in the lower toolbar row, select the script you want to run.

2 Run it

Clicking the ► **Run** button runs the selected script on the current screen.

3 Operate as needed

Choose one of the following three options during execution.

- **Just wait:** Please wait for the script to finish.
- **Pause:** Click || **Pause**. Clicking it again resumes from the same position.
- **Stop:** Click ■ **Stop**. Execution is aborted. It cannot be resumed.



About Input While AUTOTYP Is Running

While AUTOTYP is running, keyboard and mouse input to the 5250 screen is temporarily disabled. To enter input, first perform the "Pause" or "Stop" described above, and then operate.

5.2 Automatic Execution After Connecting

If you set up a "post-connection automatic-execution AUTOTYP" for each connection configuration, simply connecting can automatically perform everything from signing on to displaying the menu. For how to set this up, see the "Administrator Guide".

5.3 Recording Operations

You can automatically record keyboard and mouse operations as an AUTOTYP script.

1 Start recording

Click the ● **Rec** button. The button turns red to indicate that recording is in progress.

2 Perform the operations to record

Perform the operations you want to save as AUTOTYP, such as key input, function keys, and pasting.

3 Stop recording

Clicking the ● **Rec** button again stops recording and opens the edit dialog. Review the content, modify it as needed, and then save.

5.4 Display Switching

You can change the screen display to your preference using the toolbar icons.

Icon	Function
◆	Highlights input fields in yellow. You can see at a glance where input is possible (on / off).
/	Switches whether column separators are shown as vertical lines or as dotted lines.
A / ｱ	Switches the SBCS display characters toward lowercase Latin or toward Katakana (DBCS sessions only).
{ } / ．．	Shows or hides shift characters (SO/SI) (DBCS sessions only).

The above four display switches are saved as a single set of default values for the entire viewer (= not per connection configuration). The settings are retained even after reloading the browser.

5.5 Voice Input

Using the microphone button on the toolbar, you can convert what you say into the microphone into text and send it as input to Claude. You can convey screen operations and instructions by voice, without using the keyboard.

The speech recognition language follows the viewer's display language: it recognizes Japanese (ja-JP) when the display language is Japanese, and English (en-US) when it is English (switched with the language button).

Before Using Voice Input

Voice input is available only when you use this viewer together with the **Claude extension** in **Visual Studio Code**. In addition, you need to **add a hook** to pass the voice to Claude. For the setup procedure, see the separate "Administrator Guide".

1 Start voice input

Click the microphone button. The button changes state and the microphone becomes active (the first time, the browser asks for permission to use the microphone).

2

Speak

Speak what you want to enter. What you say is converted into text.

3

Sent to Claude

The converted text is sent as input to Claude. Voice input ends automatically when you finish speaking (to stop it midway, click the microphone button again).

5.6 Online Help

Click the **Help (i)** button on the second toolbar row to open the help in a normal browser window. The first page shows the following information.

Item	Description
Version	The product version and build date. Please include them when contacting support.
License	Shows the End-User License Agreement (EULA) and the third-party notices for open-source software included in the product.
Help	Shows the User Guide (this document) and the Admin Guide.

When you open a guide, the screen is divided into two panes.

Pane	Description
Left — contents and search	Contains the document selector, the table of contents, and a search box. Clicking an entry in the table of contents scrolls the right pane to that heading. Typing two or more characters in the search box lists the matching places in the text; clicking one jumps to it and highlights the matched words.
Right — document	The body of the selected document, including images.



Help display language

The help opens in the viewer's display language (the language toggle button): the Japanese guides when set to JA, and the English guides when set to US.

Chapter 6

Troubleshooting

Guides you through frequently asked questions and how to check them.

6.1 Cannot Connect

Symptom	Please Check
The screen does not appear even after selecting a connection	- Check with your administrator whether IBM i is running. - Check the state of the network (LAN cable, Wi-Fi, VPN). - Reload the browser (Ctrl + R) and then try again.
COMM659 appears in the OIA	Communication has been disconnected. Click "🔌 Connect" on the toolbar to reconnect. The display disappears automatically when communication is restored.

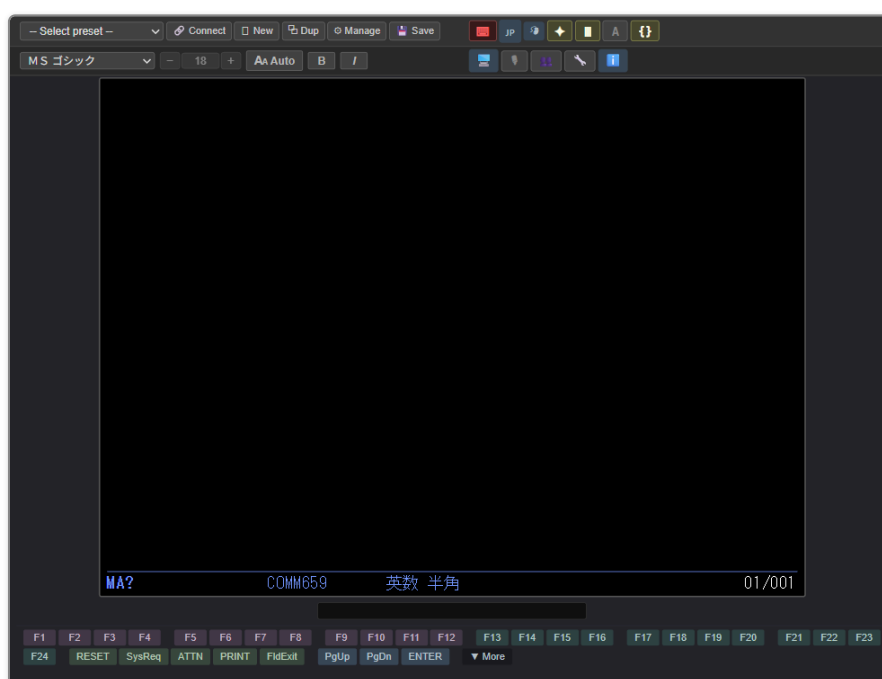


Figure 6.1-1 COMM659 display when disconnected

6.2 The Screen Does Not Respond

Symptom	Please Check
No response when pressing keys / X II appears in the OIA	This product is processing the previous operation (such as printing). Pressing the RESET key releases it.
X SYSTEM stays displayed in the OIA	The server is being awaited for a response. Normally just wait a moment. If no response returns for a long time, press SysReq (System Request), type 2 on the displayed input line, and press Enter to force-end the program or command running on the IBM i. If that still does not resolve it, consult your administrator.

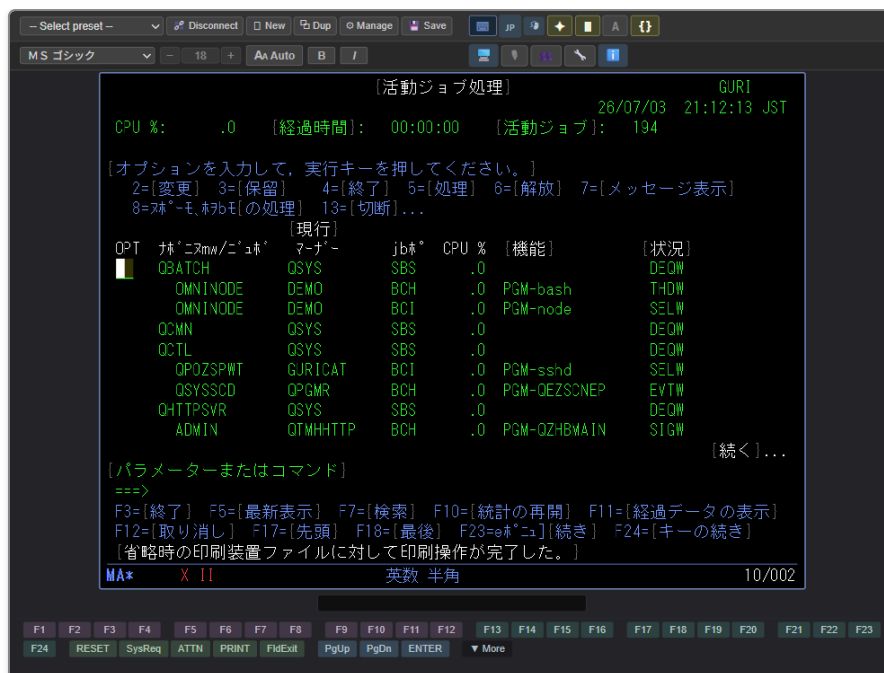


Figure 6.2-1 Example of X II appearing right after the print key (printing in progress)

6.3 Other

Symptom	Please Check
The characters are too small to see	Use the + button in the second toolbar row to increase the font size. Setting it to "Aa Auto" automatically adjusts the size to fit the window.
Keyboard input is not reflected on the 5250 screen	It is set to KBD OFF. Click the KBD OFF button or press Ctrl + K to turn it back ON.
Cannot enter Japanese	Check that the IME is enabled. Check whether the OIA display is stuck on "Alphanumeric".

Chapter 7

Glossary

Explains the technical terms used in this guide.

IBM i

An operating system for core enterprise systems provided by IBM. It was formerly called AS/400 and iSeries.

5250

The name of the terminal protocol used for communication with IBM i. This product implements TN5250E (5250 over Telnet).

Sign-on

Refers to logging in to IBM i by entering a user name and password.

Function key

Keys such as   in the top row of the keyboard. In 5250 applications,   are used.

OIA (Operator Information Area)

An abbreviation for Operator Information Area. Refers to the status line displayed at the bottom edge of the screen, showing the keyboard state and so on.

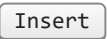
X II / X SYSTEM

OIA displays indicating that the keyboard is locked. X II indicates that this product is performing processing, and X SYSTEM indicates that it is waiting for a response from the server.

COMM659

An OIA display indicating that communication has been disconnected. It disappears automatically when communication is restored.

Insert mode

A mode in which entered characters are inserted in between at the cursor position. Switch it with the  key. In insert mode, "^" appears at OIA column 52.

AUTOTYP

This product's own mechanism for automatically running a previously recorded sequence of input operations.

MCP (Model Context Protocol)

A communication protocol for invoking external tools from generative AI agents such as Claude. This product works as an MCP server, allowing AI agents to automatically operate 5250 screens.

Connection configuration

A setting that bundles together the host name, port number, code page, and so on of a connection destination. You can switch between and use multiple connection destinations.

DBCS / SBCS

DBCS refers to double-byte characters (kanji, hiragana, katakana, etc.), and SBCS refers to single-byte characters (alphanumerics, half-width katakana, etc.).

Chapter 8

Keyboard Reference

A list of the keyboard shortcuts available in this product.

8.1 Basic Operation Keys

Key	Function
	Sends the entered content.
	Moves to the next input field.
+	Moves to the previous input field.
	Moves to the first input field.
	Moves the cursor up, down, left, and right.
	Moves the cursor one character to the left (does not delete the character).
	Deletes the character at the cursor position and shifts the text to the right leftward.
	Switches between insert mode and overwrite mode (on / off).
	Erases from the cursor position to the end of the line (Erase EOF).
	Used for closing dialogs, sending the ATTN key, and so on.

8.2 Function Keys

Key	Function
–	Sent as F1–F12 as is.
+ –	Sent as F13–F24.

8.3 Clipboard Operations

Key	Function
+	Copies the selected range to the clipboard.
+	Pastes the contents of the clipboard.
+	Pressing this immediately after pasting undoes the paste.
Mouse drag	Selects a range (no modifier key needed).
+ arrow keys	Moves the cursor to select a range (a selection method that does not use the mouse).

8.4 Special Operations

Key	Function
+	Switches the keyboard input target (KBD ON / OFF).
Left (alone)	Sends the RESET key.
Right (alone)	Sends Field Exit.

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For inquiries, contact your in-house systems staff